

Emerging radar-based technologies for cuffless blood pressure monitoring—a systematic review



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Hypertension is the leading risk factor for cardiovascular disease, the most common cause of death worldwide. Less than half the people with high blood pressure are aware of their diagnosis, and only a fifth are adequately treated. Because symptoms are often absent, hypertension might be diagnosed only after irreversible end-organ damage. Cuff-based blood pressure measurement devices provide only intermittent monitoring and can be inaccurate in some individuals. Tackling the limitations of existing devices requires improved diagnostic tools. Radar-based devices have been proposed as alternatives to cuff-based devices as radar-based devices are safe, unobtrusive, and have the potential for continuous monitoring. Radar has been used to measure pulse transit time and perform pulse wave analysis in humans, allowing blood pressure to be tracked or predicted. This Review summarises published works comparing the accuracy of radar-based blood pressure measurement devices with that of gold-standard cuff-based devices. Additionally, the Review highlights the advantages and challenges of translating this technology into clinical practice. The search was performed using PubMed, MEDLINE, IEEE Xplore, Google Scholar, Cochrane Library, Web of Science, medRxiv and bioRxiv to identify relevant publications from January 1, 1990 up to August 31, 2024. This Review identified 23 articles and found that early feasibility studies indicate that radar-based blood pressure estimation is technically feasible and can achieve promising accuracy under controlled conditions. However, because of the use of heterogeneous methodologies, small sample sizes, and inclusion of narrow blood pressure ranges, the current evidence remains constrained. Clinical translation of this technology will require evaluation in large, multimorbid cohorts, including people with hypertension. In addition, the development of universally accepted validation protocols by authoritative bodies will be essential to ensure that new devices meet rigorous performance criteria before commercial introduction, thereby establishing their credibility.

Introduction

High blood pressure is the leading risk factor for cardiovascular disease,¹ the most common cause of death worldwide.² In the UK, hypertension costs the National Health Service an estimated £2.1 billion per year and is the leading cause of heart attacks and strokes.³ A key challenge in the diagnosis of hypertension lies in the absence of symptoms until the development of associated cardiovascular disease, allowing high blood pressure to remain unnoticed for decades. 46% of individuals with hypertension are unaware of their condition and less than half of them (21%) only are adequately treated.⁴ Owing to the absence of symptoms, regular monitoring using an accurate device is the most effective way to detect hypertension before the onset of irreversible pathology.

The mainstay of measuring blood pressure remains cuff-based techniques, either with a manually inflated cuff and sphygmomanometer or with an oscillometric device. Manual auscultation is the gold-standard for validating new devices⁵ as it is accurate and reproducible. However, the manual auscultation technique requires training, can be difficult in noisy environments, and depends on the auditory acuity of the user.⁶ Therefore, manual auscultation is no longer performed by health-care professionals frequently. The most common blood pressure devices used in hospitals, clinics, and homes are oscillometric cuffs. These devices are inflated, usually around the upper arm, by an electronic pump until the arterial pulse is occluded and

then deflated. During deflation (or less commonly inflation), the machine detects a series of pulse waves that vary in magnitude as the cuff pressure surpasses and then falls below mean arterial pressure. Oscillometric machines use various algorithms that combine changes in pulse wave volume with their rate of change to calculate systolic blood pressure (SBP) and diastolic blood pressure (DBP). The algorithm differs among devices; hence, as per international guidelines, every new machine should be validated against a sphygmomanometer.⁵ When the pulse is irregular, as during atrial fibrillation, or if the arm moves under the cuff, the pulse wave volume does not change smoothly, thus making the estimation of blood pressure by the machine difficult. In people with marked bradycardia, too few pressure waves might be detected hindering the accurate functioning of the algorithm.⁶ Therefore, oscillometric machines can be inaccurate in such individuals. Additionally, cuff-based methods can cause bruising, spread contact-based infections, and do not allow continuous monitoring. Although cuff-based devices have been invaluable in addressing hypertension to date, the high proportion of undiagnosed and uncontrolled hypertension cases indicates the need for novel diagnostic tools to bridge the gap left by current devices.

Alternative approaches for blood pressure monitoring have been investigated extensively since the beginning of the 21st century (table 1), with focus on cuffless methods, including radar. The term radar was coined in 1940 as an

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	Description	Advantages	Disadvantages
Invasive method			
Intra-arterial estimation	A catheter is inserted into an artery via an arterial puncture and connected to a pressure transducer	Provides direct intra-arterial pressure measurement and is the most accurate method; performs continuous monitoring	Arterial puncture risks bleeding; in-situ catheter risks thrombosis and infection; unsuitable outside critical care setting; systolic pressure varies with the location of measurement (eg, radial, brachial, or aortic) owing to wave reflection in the circulation
Cuff-based methods			
Manual auscultation	A cuff is manually inflated around the upper arm and deflated while a physician listens for the Korotkoff sounds, typically over the brachial artery using a stethoscope; the first sound (K1) is the systolic blood pressure and the final sound (K5) is the diastolic blood pressure	Well established and reproducible	Requires training; requires quiet environment and adequate hearing; does not allow continuous or contactless monitoring; cannot be performed at home; can be uncomfortable; no option to perform central aortic blood pressure measurement
Oscillometric cuff-based estimation	A cuff is electronically inflated and deflated around the upper arm or wrist while sensors within the cuff are used to detect the change in pulse rate and volume; this information is used to calculate blood pressure using algorithms	Does not require training; commonly available; low-cost devices	Inaccurate in some patient groups (patients with atrial fibrillation, bradycardia, or ectopy); does not allow continuous or contactless monitoring; can be uncomfortable
Volume clamping (Penaz method)	A cuff with integrated PPG is placed around the finger; light absorption is kept constant by maintaining the same volume of blood in the finger by changing the pressure in the cuff, which is translated into blood pressure values	Allows continuous monitoring	Less accurate in people with peripheral arterial disease, hypotension, on vasopressors, or in a cold ambient environment; can be uncomfortable because of cuff inflation
Cuffless methods			
Applanation tonometry	A strain gauge pressure sensor placed over an artery is used to detect vessel pulsations; blood pressure is derived from features of the pulse using mathematical formulas or by calibration using cuff pressures	Allows continuous monitoring	Continuous pressure on the artery can be uncomfortable; highly sensitive to misplacement of probe or underlying movement; usually requires calibration with cuff pressure
Contactless distension wave analysis	A sensor is directed at an artery to monitor distension wave changes; distension wave parameters that relate to blood pressure are used to calculate blood pressure based on training data; distension wave can be detected using PPG-based or radar-based sensors	Can be built into a wearable device (such as a smartwatch); allows continuous monitoring; comfortable for user	Highly sensitive to misplacement if not positioned correctly over the pulse; relies on training data to calibrate baseline distension wave
Estimation of blood pressure through measurement of pulse transit time and PWV	Sensors are used to detect two pulses, and the speed at which blood travels between the points is measured to track blood pressure; measurements can be made using a variety or combination of sensing techniques such as electrocardiography, PPG, video PPG, ultrasound, bioimpedance, radar, and mechanical signals such as ballistocardiogram or mechanocardiogram	Allows continuous monitoring; comfortable for user	Requires accurate calibration using cuff-based methods at baseline and at regular intervals thereafter; otherwise subsequent measurements will be incorrect; usually requires two sensors so might not fit into a wearable device; PPG or video PPG are affected by light levels and skin tone

PPG=photoplethysmography. PWV=pulse wave velocity.

Table 1: Summary of methods for measuring blood pressure

acronym for radio detection and ranging, when it was used in World War 2 to detect enemy submarines and aircrafts. Measurement of vital signs using radar was first explored in 1975 to measure the respiratory rate of a rabbit and a human at a distance of 30 cm.⁷ The output signal was coupled to an oscilloscope and charted using a strip chart recorder. Since the 1970s, major advances in sensor technology and signal processing, especially machine learning, have expanded its potential to include blood pressure estimation.

Radar is safe, inexpensive, and convenient, as it can be integrated into free-standing devices, enabling complete contactless monitoring, or into wearable devices such as smartwatches. Unlike video-based monitoring, radar-based techniques do not collect identifiable imaging data, avoiding associated privacy concerns. Radar can be programmed for intermittent or continuous physiological monitoring depending on clinical requirements. Furthermore, radar has been shown to measure both resting heart and

breathing rates accurately;^{8,9} thus, radar has the potential to record multiple vital signs of a single individual. These characteristics have fuelled growing interest in radar-based blood pressure monitoring. Consequently, research outputs in this field have increased sharply in the past 5 years (figure 1). Although no radar-based devices have received regulatory approval or are commercially available yet, studies have shown promising accuracy.

Broadly, radar devices estimate blood pressure either by exploiting its relationship with pulse transit time (PTT) and pulse wave velocity (PWV) (figure 2) or by performing pulse wave analysis (PWA) (figure 3). A detailed explanation of these methods and a discussion on basic radar concepts, aimed at clinicians, are provided in the appendix (pp 1–3).

Previous reviews have focused on the use of radar to measure pulse and breathing rates;^{8,9} however, no review has summarised the studies using radar for blood pressure measurement. This Review therefore addresses this

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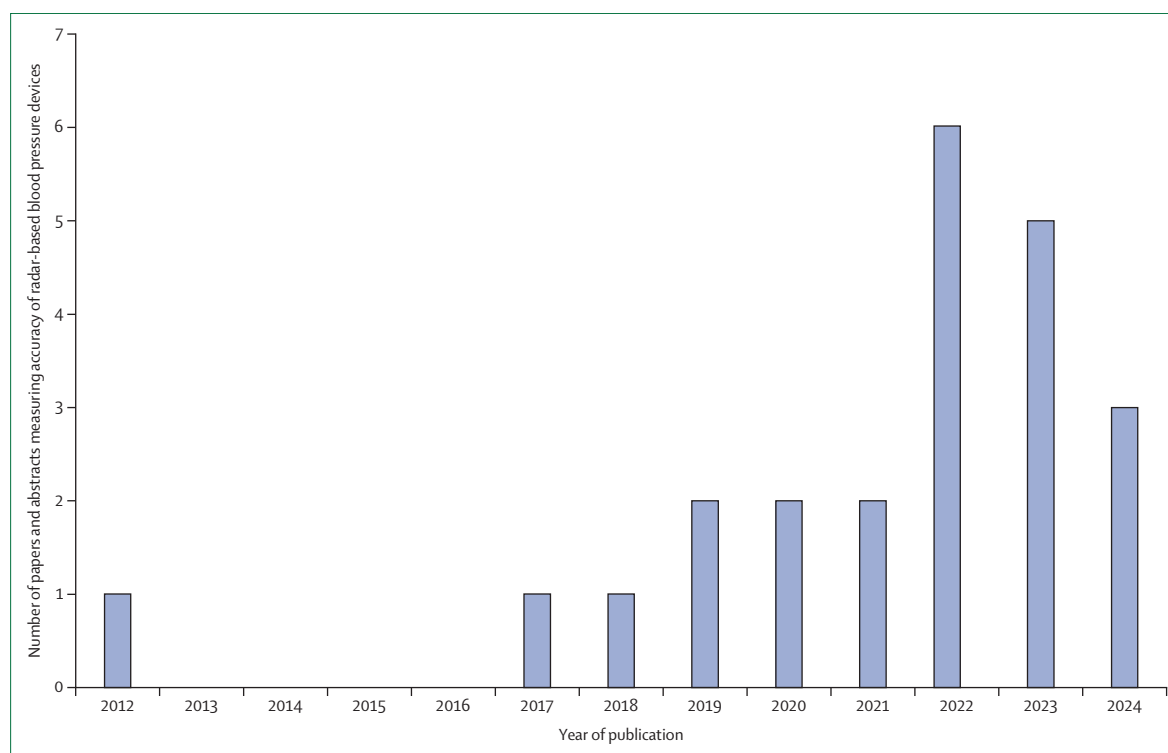


Figure 1: Histogram showing the number of papers and conference abstracts identified in this Review (n=23) published per year (January, 2012–August, 2024) assessing the accuracy of radar-based blood pressure devices

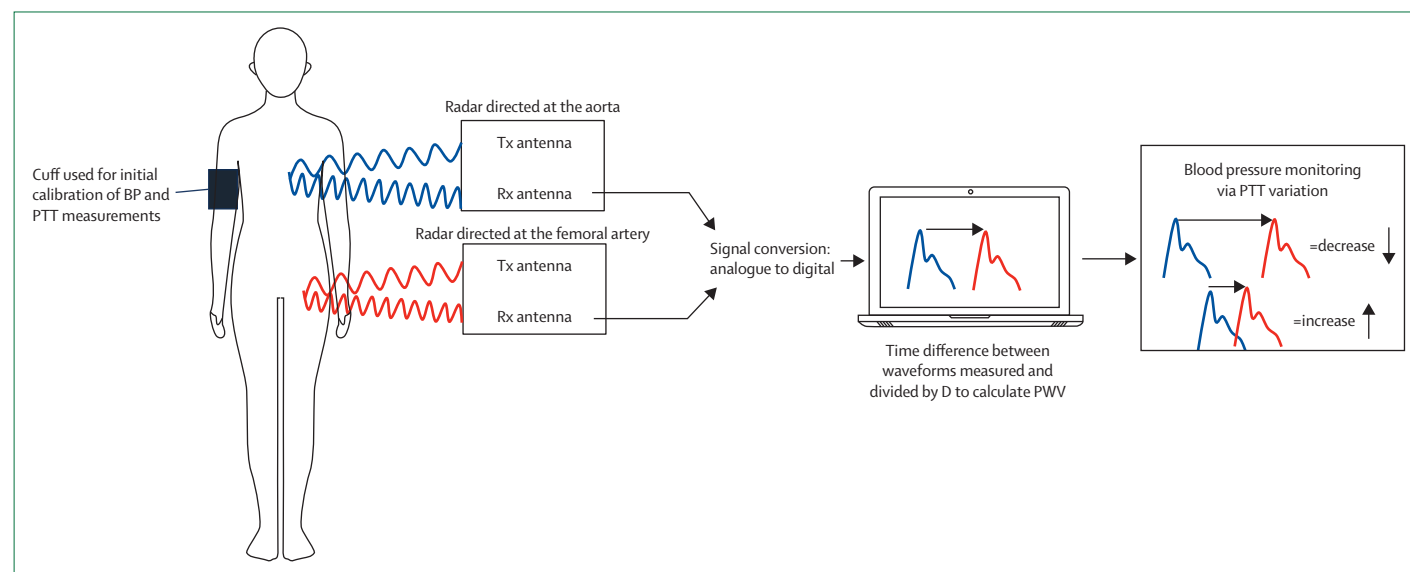


Figure 2: Representative setup for estimating blood pressure using pulse transit time (PTT)

Radar detects arterial pulses by detecting micro-displacements on the skin surface. PTT, defined as the time for a pulse to propagate between two arterial points, enables calculation of pulse wave velocity, which is inversely related to blood pressure. Tracking PTT helps to monitor blood pressure changes from a calibrated baseline established with a cuff. BP=blood pressure. D=distance. Tx=transmission. PWV=pulse wave velocity. Rx=receiving.

knowledge gap by summarising the studies that compare radar devices with cuff-based systems and by discussing potential advantages and challenges of translating this technology into clinical practice.

Methods

Search strategy and selection criteria

The review was conducted according to the PRISMA guidelines. The Review aimed to examine whether radar-

For more on PRISMA reporting guidelines see <https://www.prisma-statement.org/>

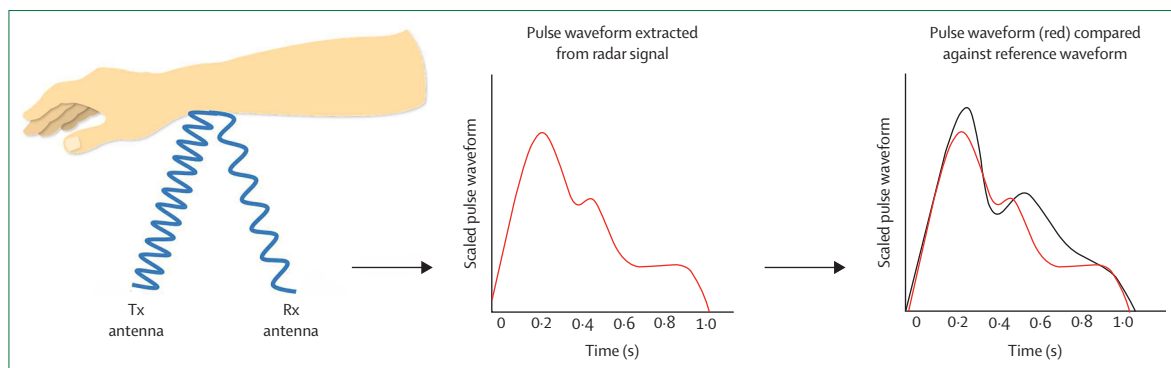


Figure 3: Schematic showing the workflow for inferring blood pressure from radial pulse wave analysis (red trace=reflected pulse waveform; black trace=reference pulse waveform)

Radar transmitted to the skin can detect the waveform of the pulse, and its shape can be extracted from the reflected signal. Using this shape, waveform features can be measured and compared with a known reference waveform, allowing blood pressure to be predicted. Tx=transmission. Rx=receiving.

based methods can be used to estimate blood pressure in humans as an alternative to standard cuff-based devices. Original research articles or preprint manuscripts and peer-reviewed conference abstracts and papers, which reported the comparison of blood pressure estimated using radar devices with that estimated using a reference device, were included. Non-English publications and reviews were excluded, but titles in the reference sections of reviews were searched manually for additional inclusion material and included if relevant. The search was performed by reviewing electronic indexing on PubMed, Google Scholar, MEDLINE, Institute of Electrical and Electronics Engineers (IEEE) Xplore Cochrane Library, and Web of Science. medRxiv and bioRxiv were also searched. All evidence included in this Review is publicly available.

A systematic search of the databases was performed to identify relevant eligible publications from January 1, 1990 up to August 31, 2024 (figure 4). Search terms are listed in the appendix (p 4). Titles and abstracts were initially screened for suitability by one reviewer (DF). If abstracts were insufficient to decide whether the publication should be included or not, the full text was read to inform the final decision, and a second reviewer (GC) was consulted in case of uncertainty. To reduce the risk of bias, all included studies and extracted data were reviewed by both the independent reviewers (DF, GC). There were no discordant rulings requiring mediation by the designated third reviewer (BB). Extracted data included year and country of study, number of participants, participant age and sex, experimental design and equipment, reference device, and the outcome measure of accuracy. Publications that did not compare blood pressure estimated using radar-based methods with that estimated using a reference device, or those that omitted a comprehensive explanation of how blood pressure was measured, were excluded. Owing to the absence of validated tools to assess study quality in this specific context, no formal quality assessment was performed. The synthesis of results is narratively described.

Results

Studies in which blood pressure measured by radar was compared with that measured using reference devices

We identified 23 articles, including journal manuscripts and conference papers, after screening and exclusion (table 2). The PTT method was used in 11 of the 23 studies,^{10–13,15–17,19,20,22,29} PWA (either from the heart or arterial pulses) was used in ten studies,^{14,18,21,24,25,27,28,30–32} and both the methods were used in two studies.^{23,26} All studies cited in this Review are presented in the appendix (pp 5–7).

Studies in which blood pressure was estimated using the PTT or PWV method

PTT was measured using peripheral pulses, central pulses, the heartbeat, or combinations thereof. Liao and colleagues¹⁷ performed the largest study, involving 73 healthy participants aged 20–67 years, in which reflective PTT-derived blood pressure from the radial artery was compared with that from a volume clamping finger cuff, initially calibrated using an oscillometric cuff. 1095 paired samples were analysed from a 10-min recording for each individual. The mean difference $\pm$ SD was 1.70 ± 5.59 mm Hg for SBP and 2.85 ± 5.57 mm Hg for DBP.¹⁷ However, of the 120 participants originally recruited, over a third were excluded from the final analysis because of poor signal quality or incomplete waveform extraction during blood pressure measurement. The blood pressure range for this cohort was narrow, and only 7 (9%) of 73 individuals recruited had an SBP exceeding 140 mm Hg. Thus, although the accuracy of blood pressure measurement was promising for the participants in whom the radar-based device was able to detect a signal, the measurement was possible in only 73 (61%) of 120 participants in the overall group. A wrist-based reflective PTT model was used in a feasibility study by Tseng and Tseng involving one healthy female participant, who underwent daily measurements for 8 days.¹⁶ The mean difference $\pm$ SD was 0.55 ± 5.45 mm Hg for SBP and 2.26 ± 3.93 mm Hg for DBP compared with those measured using an oscillometric cuff-based device.

Earlier work by Lin and colleagues compared radar-derived blood pressure from the radial artery with that estimated using an oscillometric cuff in five participants.¹⁰ The mean difference $\pm$ SD was 4.09 ± 1.69 mm Hg for SBP and 5.18 ± 1.61 mm Hg for DBP. The reduced accuracy might be attributed to the lower radar frequency (300 MHz) than that used in the studies by Tseng and Tseng and Liao and colleagues.^{16,17}

Wen and colleagues compared reflective pulse transit time from the brachial artery of seven participants with invasive intra-arterial recordings over a period of 2 min.¹⁹ The first 50 pulses were used as training data for a linear regression model, and measurements thereafter were used as test data. For all participants, the absolute error was less than 3 mm Hg compared with the reference value. The invasive nature of intra-arterial measurements limits large studies from following validation protocols in terms of ethics and feasibility. Additionally, as cuff-based and intra-arterial blood pressure values differ, validation against an invasive intra-arterial method implies that the evidence-based thresholds governing hypertension diagnosis and treatment no longer apply, thus restricting the clinical value of the results.

Geng and colleagues used two radar modules stationed 40 cm from the neck and chest to measure the aortic and carotid artery pulse; the estimated values were compared with blood pressure estimated using a volume clamp finger cuff in 27 healthy participants.²⁹ Central arteries were used to minimise the influence of smooth muscle contraction.³³ However, the accuracy and precision of the estimated blood pressure values were lower than those from other studies: mean difference $\pm$ SD for SBP was 5.54 ± 7.62 mm Hg and that for DBP was 4.68 ± 6.15 mm Hg. The large difference in values could be the result of a greater antenna-body distance, which increases the likelihood of noise, or because the radar signal was not directed at the optimal pulsation. Furthermore, this experimental set up required participants to be supine on a bed, thus restricting its widespread usability. Ishizaka and colleagues recognised that radar signals directed at the chest will be affected by noise from respiration; therefore, they proposed a deep learning model to clean the aortic pulse wave for improved peak detection.²² When recorded at a distance of 50 cm, the mean absolute error (MAE) of SBP compared with that measured using a sphygmomanometer was 3.06 mm Hg with the model applied and 4.01 mm Hg without.

A PhysioChair has inbuilt radar in the back (to target the heartbeat) and armrest (to target the radial artery). This hardware set up restricts blood pressure measurement to the seated position, thus hindering continuous ambulatory patient monitoring; nevertheless, such a set up might be useful in a ward setting where patients are highly sedentary.²⁰ Mean relative error of SBP and DBP readings in seven participants was 1.99%. Other vital signs including respiratory rate and heart rate were recorded simultaneously. Subject to validation, the PhysioChair could relieve health-care workers of a major workload by reducing the time spent on vital sign measurement; nevertheless, the

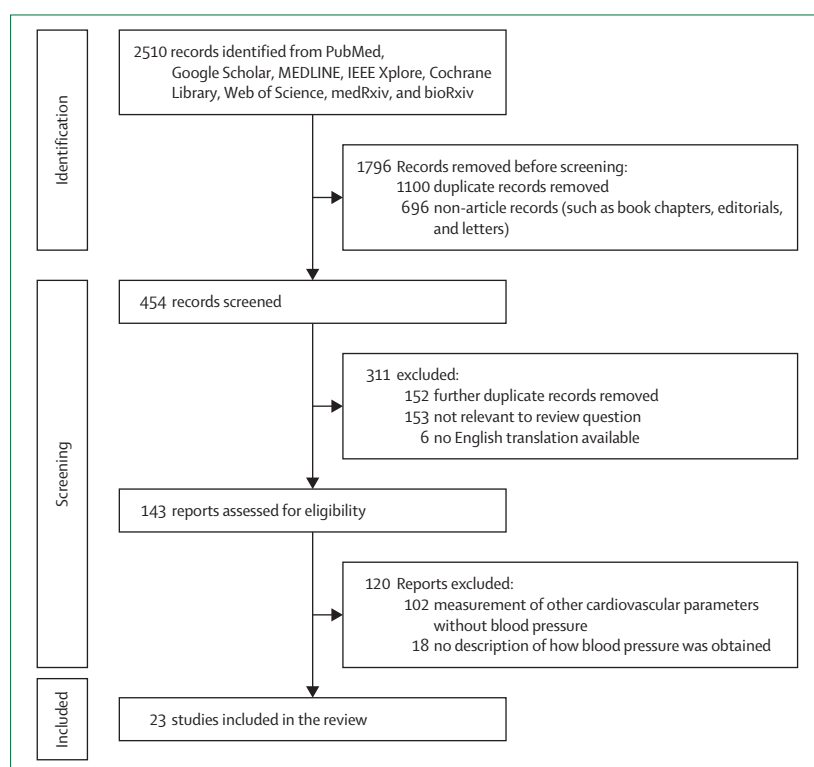


Figure 4: PRISMA flow chart detailing identification of studies via databases and registers
IEEE-Institute of Electrical and Electronics Engineers.

time for disinfecting the chair between individual use would need to be factored in.

Radar has been combined with other modalities such as bioimpedance,^{11,15} photoplethysmography,^{13,15} and electrocardiography¹³ for proximal pulse detection. Buxi and colleagues reported moderate correlations for SBP ($r=-0.66$) and DBP ($r=-0.4$),¹¹ whereas Pour Ebrahim and colleagues found that approximately 35 (81%) of 43 SBP readings fell within 7 mm Hg of reference sphygmomanometric measurements.¹³ Heydari and colleagues combined radar with bioimpedance or photoplethysmography in various postures, during exercise, and after glyceryl trinitrate administration.¹⁵ The radar-bioimpedance model provided readings with the highest accuracy compared with the readings using an oscillometric cuff. At rest, 96% of SBP and 97% of DBP measurements were within 10 mm Hg; after exercise, 92% of SBP and 96% of DBP measurements were within this range; and after glyceryl trinitrate administration, 96% of SBP and 100% of DBP measurements were within 10 mm Hg (no mean difference or MAE reported).¹⁵ However, these multisensory devices are becoming less popular owing to high cost, complexity, and user inconvenience.

Studies in which blood pressure was estimated using the PWA method

Vysotskaya and colleagues positioned a radar 3 mm above the radial artery in 55 participants. In the initial work from

	Country of study	n	Age, * years (mean ± SD or range)	Proportion of female participants (n, %)	Health status of the cohort	Radar type; transmission frequency	Positioning of radar-based device	Other sensors used	Source of baseline blood pressure data estimation	Method used to estimate blood pressure	Position and setting	Reference device	Result for SBP (mean ± SD unless stated, mm Hg)	Result for DBP (mean ± SD unless stated, mm Hg)
Lin et al (2012) ^{10†}	Taiwan	5	33-5	0	Healthy	UWB; 300 MHz	On wrist	None	Calibration BP for model; same individual	PWV from PTT (peripheral arteries)	At rest; research laboratory	Oscillometric cuff	4.09 ± 1.69	5.18 ± 1.61
Buxi et al (2017) ¹¹	Australia	6	NA	0	Healthy	CW; 1.1 GHz	On chest; third intercostal space	Blmp and ECG	Calibration BP for model; same individual	PWV from PAT (central artery)	Rest and after exercise (bike ergometer); research laboratory	Oscillometric cuff	PAT: SBP PCC -0.66 p=0.056	PAT: DBP PCC -0.40 p=0.025
Zhao et al (2018) ^{12†}	China	3	23 ± 2	3 (66.6%)	Healthy	CW; 100 MHz	50 cm from abdomen	None	Calibration cuff BP for model; same individual	PWV from PTT (single aortic pulse wave)	At rest, supine; research laboratory	Sphyg	Absolute error <3 mm Hg all individual, SBP and DBP	Absolute error <3 mm Hg all individual, SBP and DBP
Pour Ebrahim et al (2019) ¹³	Australia	43	40-65	21 (48.8%)	Healthy	CW; 900 MHz	On sternum	ECG and PPG on earlobe	Calibration cuff BP for model; same individual	PWV from PTT (peripheral artery)	Sit, stand supine, exercise; research laboratory	Sphyg	% within 7 mm Hg rest 81; exercise 76	DBP not provided
Ohata et al (2019) ^{14†}	Japan	4	20-29	0	Healthy	24 GHz	10 cm from chest	None	Training data; same individual	Analysis of cardiac motion	At rest, seated; research laboratory	Oscillometric cuff	Ref estimate SBP PCC 0.79	Ref estimate DBP PCC 0.88
Heydari et al (2020) ¹⁵	Australia	43	52 ± 11	21 (48.8%)	NA	CW; 900 MHz	On sternum	Blmp or PPG on earlobe	Calibration BP for model; same individual	PWV from PTT (central artery)	Sitting, standing, supine, during exercise, and after glyceryl trinitrate administration; research laboratory	Oscillometric cuff and volume clamping finger cuff	% within 10 mm Hg sitting 96; exercise 92; glyceryl trinitrate administration 96	% within 10 mm Hg sitting 97; exercise 96; glyceryl trinitrate administration 100
Tseng and Tseng (2020) ^{16†}	Taiwan	1	23 ± 0	1 (100%)	Healthy	CW; 24 GHz	3 cm from wrist	None	Calibration BP for model; same individual	PWV from R-PTT (peripheral)	At rest, seated; research laboratory over 8 days	Oscillometric cuff	0.55 ± 5.45	2.26 ± 3.93
Liao et al (2021) ^{17†}	USA	73	40 ± 10	20 (27.4%)	Healthy	FMCW; 57-63 GHz	On wrist, above radial artery	None	Calibration BP for model; same individual	PWV from R-PTT (peripheral)	At rest, seated; research laboratory	Volume clamping finger cuff	1.70 ± 5.59	2.85 ± 5.57
Jung et al (2021) ^{18†}	Germany	8	32	4 (12.5%)	Healthy	CW; 300 GHz	1 m from chest	None	Training data; same individual	Analysis of cardiac motion	At rest, seated; research laboratory	Sphyg	MRE 8.04%	MRE 8.30%
Wen et al (2022) ^{19†}	China	7	NA	NA	Known cardiovascular disease	CW; 120 GHz	On elbow	None	Training data; same individual	PWV from R-PTT	At rest, seated 2 min; research laboratory	Intra-arterial	Absolute error <3 mm Hg all subjects, SBP and DBP	Absolute error <3 mm Hg all subjects, SBP and DBP
Wen et al (2022) ^{20†}	China	7	18-40	...	Healthy (4), diagnosed with hypertension (1), and diagnosed with hypotension (2)	CW; 5.8 and 120 GHz	Embedded in chair; back and armrest	None	Training data; same individual	PWV from PTT (central and peripheral)	At rest, seated; research laboratory	Sphyg	MRE SBP and DBP: 1.99%	MRE SBP and DBP: 1.99%
Kawasaki and Kajiwara (2022) ^{21†}	Japan	3	NA	NA	NA	FMCW; 79 GHz	1 m from chest	None	Training data; same individual	Analysis of cardiac motion	At rest, seated; research laboratory	Oscillometric cuff	Waveform feature; BP PCC 0.74-0.91	DBP not provided

(Table 2 continues on next page)

	Country of study	n	Age,* years (mean ± SD or range)	Proportion of female participants (n, %)	Health status of the cohort	Radar type; transmission frequency	Positioning of radar-based device	Other sensors used	Source of baseline blood pressure data estimation	Method used to estimate blood pressure	Position and setting	Reference device	Result for SBP (mean ± SD unless stated, mm Hg)	Result for DBP (mean ± SD unless stated, mm Hg)
(Continued from previous page)														
Ishizaka et al (2021) ²²	Japan	6	NA	NA	NA	CW; 24 GHz	50 cm from chest	None	Training data; same and new individuals	PWV from R-PTT (central artery)	At rest, seated; research laboratory	Sphyg	MAE 3·06	DBP not provided
Zheng et al (2022) ^{23†}	China	27	22–27	NA	NA	CW; both 2·4 GHz	Above the chest and below the wrist	None	Training data; same and new individuals	PWV from PTT and pulse wave analysis	At rest, supine and post exercise; research laboratory	Sphyg	MAE 9·51 ± 12·13	MAE 6·69 ± 8·12
Shi et al (2022) ^{24†}	Singapore	3	NA	1 (33·3%)	Healthy	FMCW; 24 GHz	15 cm from chest	None	Training data; same individuals	Arterial pulse wave analysis	At rest, seated; research laboratory	Sphyg	2·52 ± 6·73	0·22 ± 3·85
Vysotskaya et al (2023) ²⁵	Germany	55	20–90	12 (21·8%)	Healthy	FMCW; 60 GHz	On wrist, 3 mm above the radial artery	None	Training data; same individuals	Arterial pulse wave analysis	At rest, seated; research laboratory	Volume clamping finger cuff	RMSE 9·2 ± 8·3	RMSE 7·7 ± 5·7
Singh et al (2023) ²⁶	South Korea	5	45 ± 23	2 (40·0%)	Healthy (n=4) and diagnosed with hypertension (n=1)	FMCW; 77 and 79 GHz	50 cm from chest and wrist	None	Training data; same individuals	PWV from PTT and pulse wave analysis	At rest, seated; research laboratory	Oscillometric cuff	RMSE 3·33	RMSE 3·14
Jiang et al (2023) ²⁷	China	36	NA	NA	NA	UWB; 23 GHz	60 cm from chest	None	Training data; same and new individuals	Arterial pulse wave analysis	Rest and post exercise, seated; research laboratory and simulated clinic	Sphyg	MAE 4·73 ± 5·52	MAE 4·49 ± 5·25
Shi et al (2023) ^{28†}	Singapore	24	NA	NA	Healthy (n=21), diagnosed with hypertension (n=2), and diagnosed with hypotension (n=1)	CW; 2·4 GHz	15 cm from chest	None	Training data; same individuals	Arterial pulse wave analysis	At rest, seated; research laboratory	Sphyg	MAE 9·1	MAE 7·8
Geng et al (2023) ²⁹	China	27	23–46	6 (22·2%)	Healthy	FMCW; 77–81 GHz	40 cm from neck and chest; single module	None	Training data; same individuals	PWV from PTT (central artery)	Post exercise, supine; research laboratory	Volume clamping finger cuff	5·54 ± 7·62	4·68 ± 6·15
Vysotskaya et al (2024) ^{30†}	Germany	55	NA	NA	NA	FMCW; 60 GHz	On wrist	None	Training data; same individuals and public database	Arterial pulse wave analysis	At rest, seated; research laboratory	Volume clamping finger cuff	MAE 1·02 ± 1·77	MAE 1·57 ± 2·36
Wang et al (2024) ^{31†}	China	70	25–80	40 (57·1%)	Healthy (n=57) and diagnosed with hypertension (n=13)	UWB; 7·3 GHz	30–60 cm from chest	None	Training data; different individuals and public database	Arterial pulse wave analysis	At rest, seated; research laboratory	Oscillometric cuff	MAE 6·5 ± 6·1	MAE 4·7 ± 4·9
Cao et al (2024) ^{32†}	China	35	20–54	13 (37·1%)	Healthy	FMCW; 77 GHz	30–90 cm from forearm	None	Training data	Arterial pulse wave analysis at multiple sites of arm	At rest, seated; research laboratory	Oscillometric cuff (on other arm)	MAE 2·05 ± 6·83	MAE 1·99 ± 6·30

All studies used single point-in-time measurements except the study by Tseng and Tseng,¹⁶ in which measurements were performed over 8 days. Where multiple sets of results exist (eg multiple regression models used on the same dataset or the same model on different datasets), only the best performing dataset is reported in this table. BImp=biompedance. BP=blood pressure. CW=continuous wave. DBP=diastolic blood pressure. ECG=electrocardiogram. FMWC=frequency-modulated continuous wave. GHz=gigahertz. MAE=mean absolute error. MRE=mean relative error. NA=not available. PAT=pulse arrival time. PCC=Pearson's correlation coefficient. PPG=photoplethysmography. PTT=pulse transit time. PWV=pulse wave velocity. RMSE=root mean square error. R-PTT=reflective pulse transit time. SBP=systolic blood pressure. Sphyg=sphygmomanometer. UWB=ultra-wideband. *Age is reported as mean ± SD or range depending on the availability of data in the original manuscript. †Conference paper.

Table 2: Summary of published studies that used radar for blood pressure estimation

this group, neural networks were trained using data from the same cohort as that tested. 21 distension wave features were integrated with clinicodemographic data such as height and weight to estimate blood pressure.²⁵ Compared with the measurements using a volume clamp finger cuff, radar resulted in the root mean square error $\pm$ SD of 9.2 ± 8.3 mm Hg for SBP and 7.7 ± 5.7 mm Hg for DBP. A limitation of this study, as recognised by the authors, was that the training data included relatively few female participants (12 [22%] of 55) and participants with systolic blood pressure >140 mm Hg (10 [18%] of 55). In a follow-up study from the same group, the publicly available PulseDB dataset³⁴ was used as the training data. The PulseDB dataset contains blood pressure and distension wave data from 5361 participants, including a greater range of baseline blood pressure values than that in the original study. Accuracy also improved substantially compared with that of the original study: MAE $\pm$ SD was 1.02 ± 1.77 for SBP and 1.57 ± 2.36 mm Hg for DBP.³⁰ Similarly, Wang and colleagues used a publicly available database to train a deep learning model to analyse pulse waveforms from the chest detected by radar. The model was tested on 70 participants (including individuals with hypertension) yielding an MAE $\pm$ SD of 6.5 ± 6.1 mm Hg for SPB and 4.7 ± 4.9 mm Hg for DBP, compared with the readings from an oscillometric cuff.³¹

PWA has been performed using devices positioned up to 60 cm from the chest, highlighting the potential for unobtrusive monitoring. Jiang and colleagues compared PWA-based blood pressure recordings measured using radar-based devices with sphygmomanometer measurements in 36 participants in a research office and simulated medical environment.²⁷ The MAE $\pm$ SD was 4.73 ± 5.52 mm Hg for SBP and 4.49 ± 5.25 mm Hg for DBP. Moreover, some participants in the test cohort were not part of the training cohort; thus, the algorithm was validated against unseen data. The results of that study were less accurate than those of the follow-up study by Vysotskaya and colleagues.³⁰ Nevertheless, further algorithm refinement using larger datasets might improve performance. In the initial work by Shi and colleagues, PWA-based blood pressure readings were estimated using radar-based devices positioned 15 cm from the chest in three participants. The mean difference $\pm$ SD was 2.52 ± 6.73 mm Hg for SBP and 0.22 ± 3.85 mm Hg for DBP, compared with the readings from a sphygmomanometer.²⁴ A follow-up study with a similar hardware set up in 24 participants showed an MAE of 9.1 mm Hg for SBP and of 7.8 mm Hg for DBP.²⁸ The change in accuracy could be explained by the radar frequency shift (from 24 to 2.4 GHz) in the follow-up study, inclusion of participants with diagnosed hypotension and hypertension, and the difference in signal post-processing technique. The cardiac motion waveform, rather than an arterial pulse wave, has been used to predict blood pressure at a distance of 1 m in studies with low sample sizes ($n=3$ and $n=8$ healthy participants).^{18,21} Notably, studies that used PWA from the chest required up to 20 s of breath

holding to reduce motion artifact from respiration; the breath-holding condition poses major problems for people who are frail in a clinical setting or who feel breathless. Cao and colleagues used a single radar transmitter that was beam-steered to perform PWA at multiple sites along the forearm of 35 participants.³² The varying pulse waves at each site were used to derive a blood pressure transfer function, ie, the amplitude and phase ratio between waveforms. The blood pressure transfer function describes how compliance changes along an artery and allows inference of blood pressure using machine learning. The MAE $\pm$ SD was 2.05 ± 6.83 mm Hg for SBP and 1.99 ± 6.30 mm Hg for DBP, compared with oscillometric readings. The blood pressure transfer function is widely used in cuff-based commercial devices that are used to estimate central aortic blood pressure from radial PWA, and validation studies have shown sufficient agreement between invasive and inferred blood pressure readings, following calibration using invasive techniques.³⁵

Studies in which blood pressure was estimated using both PWV and PWA methods

Singh and colleagues used two radar-based devices, one positioned 50 cm from the chest to detect aortic pulses and the other positioned 50 cm from the wrist to detect radial pulses.²⁶ Of the various distension wave features evaluated, distension wave area under the curve (AUC) correlated best with blood pressure readings. PTT was then calculated and combined with the AUC to estimate blood pressure, and the root mean square errors were 3.33 mm Hg for SBP and 3.14 mm Hg for DBP. Thus, the results were more accurate than those using PTT or AUC alone in a small study of five healthy participants. Similarly, Zheng and colleagues combined PTT with features extracted from the radial pulse wave to estimate blood pressure in a cohort of 27 participants, yielding an MAE $\pm$ SD of 9.51 ± 12.13 mm Hg for SBP and 6.69 ± 8.12 mm Hg for DBP.²³ Although root mean square error and MAE are not directly comparable, Zheng and colleagues reported larger absolute error and variability than Singh and colleagues, suggesting lower accuracy of the device, which may in part be explained by the use of lower-frequency radar (2.4 GHz vs 23 GHz). Furthermore, the participants in the cohort tested by Zheng and colleagues exercised before measurement to increase blood pressure. Thus, the algorithm could have been less effective in predicting blood pressure at higher ranges or the preceding exercise could have exacerbated respiratory motion artifacts, resulting in reduced accuracy.

Discussion

Overall, evidence from small-scale feasibility studies conducted under controlled research conditions (largely in healthy, younger participants) suggests that radar-based blood pressure estimation is technically feasible and can achieve acceptable accuracy relative to reference cuff-based methods. However, the evidence base is restricted by small sample sizes and narrow blood pressure ranges, which

limits generalisability to older and more clinically diverse populations.

Limitations of existing works

Although some promising results were obtained, radar-based blood pressure measurement technology remains in its infancy. The studies have been conducted at small scales and have been designed to highlight feasibility of a prototype rather than to validate a device for clinical use. Most studies recruited only healthy, participants predominantly aged 20–50 years old, resulting in the testing of a narrow range of blood pressure. Thus, the studies do not reflect the status of comorbid or older age populations above the age of 50, who are most in need of improved diagnosis and monitoring. The experimental environments were primarily office-based, which do not simulate clinical situations where greater noise and interference would likely reduce accuracy. All studies have been performed in a resting position, either seated or supine. Once built into a wearable device, testing during ambulation might be necessary. It remains unclear if non-cardiovascular motion could be filtered adequately to detect distension waves with lower signal-to-noise ratios that might occur during ambulation or with some conditions such as Parkinson's disease or other movement disorders. Considerable variations between experimental set-ups (eg, variations in radar frequency, positioning of radar-based device, and reference devices) and signal processing techniques challenge the direct comparison of results and obscure the optimal conditions for device accuracy.

To capture emerging data in a rapidly advancing field, we included peer-reviewed conference papers along with full texts of journal manuscripts. Although most studies included in this Review were conference proceedings (16 of 23, listed in appendix pp 5–7), the associated manuscripts were full texts published in the proceedings of reputed events such as the *IEEE Engineering in Medicine and Biology Conference* and *IEEE Conference on Computer Communications*. These conferences enforce a rigorous, blinded peer-review process, often with acceptance rates similar to or lower than those of many journals (eg, typically less than 25%). In fields such as biomedical engineering and computer science, conferences are primary publication venues and often represent the first and most complete event for the dissemination of novel research. As such, the methodological rigour and scientific contribution of these papers are on par with those of traditional journal articles. Thus, despite the publication format differing from clinical and biomedical norms, the credibility and quality of the included studies remain undiminished.

Radar safety

Although radar-based radiations are non-ionising, there are potential adverse effects on the human body from exposure to radiofrequency waves. Stimulation of nerves (resulting in pain or muscle contraction) is seen at lower frequencies,

whereas tissue heating is the main concern at higher frequencies.³⁶ As high frequencies are more commonly used for vital sign monitoring, thermal heating is the most likely issue to be encountered with radar-based radiations. The likelihood of such events depends on the amount of electromagnetic radiation absorbed, measured by the specific absorption rate (SAR). The SAR for devices positioned closer to the body, for example, on chairs, will be higher than the SAR of truly remote systems. To prevent adverse events, clear international guidelines on radar exposure limits have been issued by the International Commission on Non-Ionizing Radiation Protection stipulating SAR limits must be acknowledged by device manufacturers.³⁷ Nevertheless, the SAR for a typical radar-based blood pressure measurement device is lower than that of a smartphone. Moreover, there is no evidence that chronic exposure to radar below advised limits is related to adverse health outcomes including cancer or infertility; thus, radar would be suitable for long term or even continuous blood pressure monitoring.

Challenges in translating radar to clinical medicine

Major hurdles need to be overcome before successful deployment of radar-based devices in routine clinical practice. These challenges can broadly be classified as radar-specific concerns, general challenges of cuffless monitoring, and clinical barriers.

Radar-specific concerns: radar is highly sensitive to other sources of movement in individuals (such as a moving limb), termed random body movements.⁹ The signals from random body movements are larger than those of the distension waves and can distort the measurement. Research participants are typically healthy individuals who can follow instructions clearly, whereas more random body movements would be expected in real-world patients. Another source of signal distortion is clutter, referring to signal returned from anything other than the target.³⁸ In clinical environments, clutter might arise from caretakers, parents, or even inanimate objects. On-body devices experience less clutter than remote devices owing to less space between the device and the target. Nevertheless, the amount of clutter generated in medical settings will be greater than that in research environments. How radar systems would fare outside of such tightly controlled environments remains unclear, and more sophisticated signal processing techniques might be required to remove non-cardiovascular noise.

Jiang and colleagues noted a lower signal-to-noise ratio in female participants than in male participants when blood pressure was measured from the aorta.²⁷ Although this trend was not noted in other studies, women were under-represented in most studies and signal interference from breast tissue is a possible reason. Hence, this trend should be evaluated further. Participants with increased breast or adipose tissue might require alternative signal amplification techniques during processing to generate waveform signals suitable for interpretation.

General challenges of cuffless monitoring: owing to the growing health technology market, over 500 cuffless blood pressure measurement devices are commercially available without any robust clinical validation.³⁹ Standardised validation protocols should be established to ensure accuracy of new devices, as the availability of inaccurate devices might erode trust in contactless monitoring before the validated technique is established. Owing to the highly variable nature of cuffless devices, a one-size-fits-all approach to validation will not suffice. Therefore, there is no universally accepted validation standard for cuffless devices unlike for cuff-based blood pressure devices. The IEEE,^{40,41} European Society of Hypertension (ESH),⁴² and International Organization for Standardization (ISO)⁴³ have published separate recommendations that focus on different types of cuffless blood pressure devices (appendix p 8) and mandate different criteria.

The ESH recommendations focus on cuffless intermittent blood pressure measurement, whereas the ISO standard is tailored towards continuous monitoring. The IEEE standard is designed for wearable devices in both categories. The flexibility of radar-based hardware implies that new devices could be designed to suit intermittent or continuous monitoring. Among the recommendations from the three organisations, the ESH guideline provides the most detailed instructions including the validation tests required in different positions, experimental conditions, time of the day, the need for recalibration, and stipulation of the tightest pass requirements. The ISO standard advises validation against intra-arterial blood pressure, which is more challenging to carry out both ethically and practically, and realistically restricts a study to the critical care setting where patients already have arterial lines in place; thus, results cannot be generalised to participants outside intensive care. Therefore, among the current standards, the IEEE standard might be practical for continuous-monitoring devices and the ESH guideline might be practical for intermittent-monitoring devices. However, a universal protocol must be agreed upon to guide future validation studies as the market for these devices inevitably grows. Similarly, no protocol exists for regulatory clearance of cuffless blood pressure devices, requiring collaboration with regulatory bodies in the absence of predicate devices (appendix p 9).

As blood pressure is tracked or predicted rather than directly measured, baseline cuff-based blood pressure data remains a requirement, either via individual calibrations or large-scale training datasets. The devices requiring individual calibration necessitate that the user takes a self-measurement of their blood pressure using a cuff-based system. Therefore, user compliance in calibration technique and frequency is crucial. Further, the contactless device is only as accurate as the device against which it is calibrated, potentially inviting another layer of error if inaccurate or non-validated devices are used for calibration.⁴⁴ This scenario is complicated further as wearable

devices require calibration under different conditions, for example, during ambulation.⁴⁵ Although an initial validation test might be accurate in clinical settings, calibration of the same device at home during normal activities will be technically more challenging and might introduce additional error if handled improperly.

The accuracy of wrist-wearable photoplethysmography devices that are used to perform PWA, with and without concurrent pulse arrival time measurements, was assessed in the Microsoft-funded Aurora project using electrocardiography after one day of calibration using a cuff-based device. Regression models integrating various waveform features were tested using data from participants not included for the initial calibration and compared with another regression model using only the baseline calibration blood pressure without integrating PWA. However, the accuracy of the PWA-based regression models in predicting the next-day blood pressure was not higher than that of the models without PWA and pulse arrival time parameters. Confounding physiological variables such as stroke volume, influenced by hydration status, and smooth muscle contraction can affect waveform amplitude, distorting its use for blood pressure estimation.⁴⁶ Although the tested devices used photoplethysmography sensors, the same underlying physiological principles will affect radar-based devices, particularly those designed to detect peripheral pulses. These results encourage caution and underline the need for robust validation protocols when developing contactless devices. For example, the Aktiia Bracelet advises calibration using a cuff-based device at baseline and every 4 weeks thereafter to ensure accuracy⁴⁷ despite the results of the Aurora study suggesting that the PWA regression models are less accurate as little as 24 h after calibration, compared with performance immediately after calibration. Nonetheless, considerable research efforts are ongoing in this field. The reliance on calibration indicates that cuff-based blood pressure measurement systems are unlikely to be replaced entirely with the available contactless technology. If shown to be accurate, contactless devices will, at least initially, be used in parallel with cuff-based blood pressure monitors.

Training data can be used in place of individual calibrations. Many studies reviewed here used small-scale training datasets that often included the same participants for testing. There is a significant ($p < 0.05$) inverse correlation between the SD of the estimated blood pressure and the number of participants included for training.³⁴ The availability of large-scale curated datasets such as PulseDB will provide research groups with access to a diverse range of participants and blood pressure values, thereby increasing the diagnostic accuracy using machine learning.

Clinical barriers: many studies described in this Review used radar-based devices directed towards the chest and mandated breath holds for up to 20 s in the experimental protocol to improve signal quality when measuring aortic distension wave. In clinical practice, holding the breath

might be difficult for some groups, such as children younger than 16 years or those with chronic breathlessness or cognitive impairment. Similarly, wrist-based devices might not be suitable for people with movement disorders such as Parkinson's disease as a tremor might interfere with peripheral pulse detection.

Potential benefits of radar-based devices

Once the challenges are overcome, radar-based blood pressure technologies have several potential advantages. These advantages can be broadly divided into the general benefits of cuffless devices and benefits specifically attributable to radar-based devices.

Advantages of cuffless monitoring: wearable cuffless devices increase the possibility of continuous monitoring over years without discomfort or interference, whereas cuff-based devices can only offer intermittent monitoring. Phenotypes such as masked hypertension, abnormal nocturnal blood pressure,⁴⁸ or white coat hypertension that usually requires diagnosis through ambulatory blood pressure monitoring^{49,50} could be identified at home without the need for physician-ordered investigations. Wearable devices would empower individuals to self-monitor blood pressure and their response to treatment and, in some cases, perform self-titration of drugs based on plans pre-decided in collaboration with physicians. Such adjustments in treatment plans improved long-term blood pressure control in a randomised controlled trial.⁵¹ A validated cuffless, wearable device would track circadian blood pressure variation and assess the proportion of time during which blood pressure remains within a target range, supporting the clinicians in delivering personalised therapy. Cuffs can also act as fomites in clinical settings.⁵² If not cleaned between individual uses, cuffs can spread contact-based infections that could be avoided with contactless, cuffless devices. Other potential benefits of cuffless blood pressure monitoring have been discussed in detail in previous reviews.^{53,54}

Advantages of radar-based monitoring: although several sensor modalities exist for contactless monitoring, radar offers distinct advantages over other methods. Some sensor modalities such as electrocardiography, bioimpedance, and ultrasound require application of gel or stickers to the skin, whereas radar waves are able to penetrate obstacles such as clothes; thus, free-standing devices can perform measurements in clinical settings or at home without the need to apply sensors to the body, thus saving time and ensuring user convenience. Compared with video-based modalities that capture identifiable data, radar-based devices preserve confidentiality by collecting distension waves rather than images. Individuals with privacy concerns might feel reluctant to engage with digital health care owing to concerns over data breaches or misuse. Poor health has been associated with reduced trust in the protection of electronic data,⁵⁵ implying that those most in need of continuous monitoring are least likely to engage with digital devices. Thus,

this group might be more comfortable with radar-based devices than with other modalities.

Radar-based measurement of heart and breathing rates are possible.⁹ The inclusion of blood pressure measurement is theoretically possible with some forms of radar (specifically, ultra-wideband or frequency-modulated continuous radar) and would enable simultaneous measurement of multiple cardiovascular parameters. Radar-based sensors coupled with machine learning have been validated for the diagnosis of obstructive sleep apnoea⁵⁶ and have been investigated for blood glucose monitoring⁵⁷ and fall prevention,⁵⁸ theoretically paving the way for multisystem monitoring. Such monitoring would be more challenging with contact-based sensors such as photoplethysmography or ultrasound, given the increase in the number of sensors required and associated complexity and discomfort for the user. Radar also has the potential to monitor multiple participants simultaneously. Although this concept has not yet been investigated for blood pressure measurement, simultaneous monitoring has been shown to be possible for heart rate measurement in two participants with greater than 90% accuracy.⁵⁹ If shown feasible for blood pressure measurements, vital signs for multiple people could be measured concurrently, thereby saving time and reducing the workload of nurses and health-care assistants in ward or clinical settings.

Conclusion

Improved global diagnosis and monitoring of blood pressure are essential to reduce mortality and morbidity associated with hypertension-related complications such as heart attacks, strokes, and chronic kidney disease. One facet of tackling hypertension is the development of new diagnostic tools that circumvent the limitations of cuff-based devices. If shown to be as accurate as cuff-based devices, radar-based devices might offer a safe, inexpensive, and comfortable alternative that can be used for continuous monitoring. Small-scale feasibility studies of radar-based devices in largely healthy individuals under research conditions show promise with desirable accuracy similar to that of reference cuff-based systems. However, clinical roll-out of radar-based technology for blood pressure measurement will require studies on large multimorbid cohorts and real-world people with hypertension outside strictly controlled research environments across a wider range of blood pressure values. Development of a universally accepted validation protocol by authoritative bodies is required to ensure that new devices meet strict criteria before introduction to the commercial market, thereby confirming the credibility of such devices. Once the associated barriers are overcome, radar-based devices have the potential to become invaluable tools for ubiquitous blood pressure monitoring to safeguard public health.

Contributors

DF conceptualised the systematic review; performed literature search, data collection, analysis, and writing and editing of the manuscript; and

generated the figures. BB, MO, EM, IW, JCM, NC, ADH, and KZA reviewed and edited the manuscript. GC conceptualised the systematic review, reviewed and edited the manuscript, and supervised the project. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

KZA is the Chief Executive Officer of IOLABS-UK LIMITED. BB is an employee of IOLABS-UK LIMITED. NC receives funding from the UK Medical Research Council (MRC) for salary support in UCL and has served on patient safety advisory boards to Astra Zeneca. All other authors declare no competing interests.

Data sharing

Datasets generated from this Review are available from the corresponding author upon request.

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